

Algorithms and Probability**Theory Sheet T3****FS 2026**

The solutions for this sheet must be submitted on Moodle by 02 April, 15:59.

THE FOLLOWING EXERCISE IS WORTH 2 BONUS POINTS.

Exercise T3.1 – Probability spaces

- (a) We have three fair 6-sided dice. Each die is rolled once. Describe this setting as a probability space. That is, define its sample space (“Ergebnismenge”) Ω and the probabilities of its atomic events (“Elementarereignisse”). Let E be the event (“Ereignis”) that the sum of the three dice is at least 16. List all atomic events (“Elementarereignisse”) corresponding to E . What is the probability of E ? What is the probability that the first die shows 6, conditioned on E ?
- (b) Alice and Bob have a perfectly shuffled deck of 52 cards (4 suits and 13 values). Alice deals the first two cards to herself and the next two cards to Bob. As in (a), describe this setting as a probability space. Let A be the event that Alice has a pair (i.e. two cards with the same value (and hence different suits)). Let B be the event that Bob has a pair. Compute $\Pr[A]$, $\Pr[B]$, $\Pr[A \cap B]$, and $\Pr[A|B]$.
- (c) In her collection of ancient random number generators, Samantha has two fair coins M and N . The two coins are almost identical. The only difference is that M has “heads” on both sides, while N has “heads” on one side and “tails” on the other side. Unfortunately Samantha forgot which coin is which. Luckily she developed a method to tell them apart (simply turning both coins around would be a bit too easy): She picks one of the coins at random. Both coins have the same probability of being picked. She tosses this coin $k \geq 1$ times and writes down the results. If she gets “tails” at least once, she knows for sure that she picked N . If she gets “heads” in all k tosses, she assumes that she picked M .

Model this setting as a probability space. Let E be the event that all k tosses showed “heads”. Conditioned on E , what is the probability that Samantha picked N ?

How large does k have to be so that, conditioned on E , Samantha can be 99% sure that she picked M ?

THE FOLLOWING EXERCISE DOES NOT GIVE BONUS POINTS.

Exercise T3.2 – Conditional probability

Consider a probability space with events A , B , X , and Y .

- (a) Assume that $\Pr[A|X] > \Pr[B|X]$ and $\Pr[A|\bar{X}] > \Pr[B|\bar{X}]$. Does this necessarily imply that $\Pr[A] > \Pr[B]$? Prove it or give a counterexample.
- (b) Assume that $\Pr[A|X \cap Y] > \Pr[A|X \cap \bar{Y}]$ and $\Pr[A|\bar{X} \cap Y] > \Pr[A|\bar{X} \cap \bar{Y}]$. Does this necessarily imply that $\Pr[A|Y] > \Pr[A|\bar{Y}]$? Prove it or give a counterexample.

Hint: Before solving this exercise, you may consider the following anecdote. What does it have to do with this exercise?

In the past years the usage of AI increased in all parts of our lives. Below you find a hypothetical survey that compares exam pass rates based on whether or not students regularly solve their exercises by asking their favorite AI for a solution.

	Students not using AI		Students asking AI for solutions	
	passed	failed	passed	failed
2022	900	100	100	1
2026	1	1	750	350

For your good friend Giulia-Paula Tobler the case is clear. In both years in which the survey took place it came to the same result: the pass rate of students asking AIs for solutions to the exercise sheets was significantly better then the pass rates of the students that did not use AI. Consequently, she suggest you should better start using AI asap.

Your friends argument sounds convincing to you. However, later that day when you tell your parents Marge and Homer about your conversation, they strongly disagree. What could have caused Marge and Homer to disagree?